

Reinforcement Learning for Lander Control

AE4350: Bio-inspired Intelligence and Learning for Aerospace Applications

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by

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Abbreviations

Abbreviation	Definition
ISA	International Standard Atmosphere
...	

Symbols

Symbol	Definition	Unit
V	Velocity	[m/s]
...		
ρ	Density	[kg/m ³]
...		

1. Introduction

An introduction to the lander reinforcement learning project...

2. System Description**3. RL Controller Design****4. Results****5. Learning Effect****6. Sensitivity Analysis****7. Conclusion**

Summary of findings and potential future research directions...

A. Source Code Example

Adding source code to your report/thesis is supported with the package listings. An example can be found below. Files can be added using \lstinputlisting[language=<language>]{<filename>}.

```
1 """
2 ISA_Calculator: import the function, specify the height and it will return a
3 list in the following format: [Temperature, Density, Pressure, Speed_of_Sound].
4 Note that there is no check to see if the maximum altitude is reached.
5 """
6
7 import math
8 g0 = 9.80665
9 R = 287.0
10 layer1 = [0, 288.15, 101325.0]
11 alt = [0, 11000, 20000, 32000, 47000, 51000, 71000, 86000]
12 a = [-.0065, 0, .0010, .0028, 0, -.0028, -.0020]
13
14 def atmosphere(h):
15     for i in range(0, len(alt)-1):
16         if h >= alt[i]:
17             layer0 = layer1[:]
18             layer1[0] = min(h, alt[i+1])
19             if a[i] != 0:
20                 layer1[1] = layer0[1] + a[i]*(layer1[0]-layer0[0])
21                 layer1[2] = layer0[2] * (layer1[1]/layer0[1])**(-g0/(a[i]*R))
22             else:
23                 layer1[2] = layer0[2]*math.exp((-g0/(R*layer1[1]))*(layer1[0]-layer0[0]))
24     return [layer1[1], layer1[2]/(R*layer1[1]), layer1[2], math.sqrt(1.4*R*layer1[1])]
```

B. Task Division Example

If a task division is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.

Table 3: Distribution of the workload

Task	Student Name(s)
Summary	
Chapter 1 Introduction	
Chapter 2	
Chapter 3	
Chapter *	
Chapter * Conclusion	
Editors	
CAD and Figures	
Document Design and Layout	